20W isolated DC-DC converter DIP package,
Ultra-wide input and regulated single output



Patent Protection

3 years Warranty

CE Report EN62368-1 | UKCA Report BS EN62368-1 | CB IEC62368-1 | RoHS

FEATURES

- Ultra-wide 4:1 input voltage range
- High efficiency up to 91%
- I/O isolation test voltage 1.5k VDC
- Input under-voltage protection, output short-circuit, over-current, over–voltage protection
- Operating ambient temperature range -40℃ to +105℃
- Input reverse polarity protection available with chassis(A2S) or 35mm DIN-rail mounting(A4S) version
- Meets EN50155 railway standard

URB_YMD-20WR3 series of isolated DC-DC converter products feature an ultra-wide 4:1 input voltage with efficiencies of up to 91%, 1500VDC input to output isolation, an operating ambient temperature range of -40℃ to +105℃, input under-voltage protection, output short-circuit, over-current, over-voltage protection, which makes them widely used in industrial control, electric power, instruments, communication and railway applications. Optional packages are offered for chassis or DIN-rail mounting (A2S, A4S), adding additional input reverse polarity protection.

Selection Guide

Certification	Part No. ①	Input Voltage (VDC)		Output		Full Load Efficiency④ (%) Min./Typ.	Capacitive Load (μF)Max.
		Nominal② (Range)	Max.③	Voltage (VDC)	Current(mA) Max./Min.		
EN/BS EN/IEC	URB2403YMD-20WR3/A2S/A4S	24 (9-36)	40	3.3	5000/0	86/88	10000
	URB2403YMD-20WHR3			3.3	5000/0	86/88	10000
	URB2405YMD-20WR3/A2S/A4S			5	4000/0	88/90	10000
	URB2405YMD-20WHR3			5	4000/0	88/90	10000
--	URB2406YMD-20WR3/A2S/A4S			6	3333/0	87/89	10000
EN/BS EN/IEC	URB2412YMD-20WR3/A2S/A4S			12	1667/0	88/90	1600
	URB2412YMD-20WHR3			12	1667/0	88/90	1600
	URB2415YMD-20WR3/A2S/A4S3			15	1333/0	87/89	1000
	URB2415YMD-20WHR3			15	1333/0	87/89	1000
	URB2424YMD-20WR3/A2S/A4S			24	833/0	89/91	500
	URB2424YMD-20WHR3			24	833/0	89/91	500
	URB4803YMD-20WR3/A2S/A4S	48 (18-75)	80	3.3	5000/0	86/88	10000
	URB4805YMD-20WR3/A2S/A4S			5	4000/0	88/90	10000
	URB4805YMD-20WHR3			5	4000/0	88/90	10000
	URB4812YMD-20WR3/A2S/A4S			12	1667/0	89/91	1600
	URB4812YMD-20WHR3			12	1667/0	89/91	1600
	URB4815YMD-20WR3/A2S/A4S			15	1333/0	89/91	1000
	URB4824YMD-20WR3/A2S			24	833/0	89/91	500
	URB4824YMD-20WHR3			24	833/0	89/91	500

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Note:
① Use "H" suffix for heat sink mounting, "A2S" suffix for chassis mounting and "A4S" suffix for DIN-Rail mounting. We recommend to choose modules with a heat sink for enhanced heat dissipation and applications with extreme temperature requirements;
② The A2S and A4S Model's start-up and minimum input voltages are increased by 1VDC due to the input reverse polarity protection circuit;
③ Absolute maximum stress rating without damage (not recommended);
④ Efficiency is measured at nominal input voltage and rated output load; efficiencies for A2S and A4S Model's is decreased by 2% due to the input reverse polarity protection circuit.

Input Specifications

Item	Operating Conditions		Min.	Typ.	Max.	Unit
Input Current (full load / no-load)	24VDC nominal input series, nominal input voltage	3.3V output	--	782/30	800/50	mA
		5V output	--	926/35	947/55	
		6V output	--	936/50	958/70	
		12V output	--	926/6	947/15	
		15V output	--	916/6	937/15	
		24V output	--	916/10	937/20	
	48VDC nominal input series, nominal input voltage	3.3V output	--	391/15	400/30	
		5V output	--	463/20	474/30	
		12V output	--	458/3	469/15	
		15V output	--	458/3	469/15	
		24V output	--	458/4	469/15	
Reflected Ripple Current	Nominal input series		--	30	--	
Surge Voltage (1sec. max.)	24VDC nominal input series		-0.7	--	50	VDC
	48VDC nominal input series		-0.7	--	100	
Start-up Voltage	24VDC nominal input series		--	--	9	
	48VDC nominal input series		--	--	18	
Under-voltage Protection	24VDC nominal input series		5.5	6.5	--	
	48VDC nominal input series		12	15.5	--	
Start-up Time	Nominal input voltage & constant resistance load		--	10	--	ms
Input Filter			Pi filter			
Hot Plug			Unavailable			
Ctrl *	Module on		Ctrl pin open or pulled high (TTL 3.5-12VDC)			
	Module off		Ctrl pin pulled low to GND (0-1.2VDC)			
	Input current when off		--	2	7	mA

Note: *The Ctrl pin voltage is referenced to input GND.

Output Specifications

Item	Operating Conditions		Min.	Typ.	Max.	Unit
Voltage Accuracy	0%-100% load		--	±1	±3	%
Linear Regulation	Input voltage variation from low to high at full load		--	±0.2	±0.5	
Load Regulation	5%-100% load		--	±0.5	±1	
Transient Recovery Time	25% load step change, nominal input voltage		--	300	500	μs
Transient Response Deviation		3.3V/ 5V/ 6V output	--	±5	±8	%
		Others	--	±3	±5	
Temperature Coefficient	Full load		--	--	±0.03	%/℃
Ripple & Noise④	20MHz bandwidth, 5%-100% load		--	50	100	mV p-p
Trim	Input voltage range		90	--	110	%Vo
Over-voltage Protection			110	--	160	
Over-current Protection			110	150	190	%Io
Short-circuit Protection			Hiccup, continuous, self-recovery			

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Note:
①Under 0% -5% load conditions, ripple & noise does not exceed 5%Vo. The "parallel cable" method is used for Ripple and Noise test, please refer to DC-DC Converter Application Notes for specific information.

General Specifications

Item	Operating Conditions		Min.	Typ.	Max.	Unit
Isolation	Input-output Electric Strength Test for 1 minute with a leakage current of 1mA max.		1500	--	--	VDC
	Input/output-case Electric Strength Test for 1 minute with a leakage current of 1mA max.		1000	--	--	
Insulation Resistance	Input-output resistance at 500VDC		1000	--	--	MΩ
Isolation Capacitance	Input-output capacitance at 100kHz/0.1V		--	2000	--	pF
Operating Temperature	See Fig. 1	3.3V/ 5V /6V output	-40	--	+95	℃
		Others	-40	--	+105	
Storage Temperature			-55	--	+125	
Storage Humidity	Non-condensing		5	--	95	%RH
Pin Soldering Resistance Temperature	Soldering spot is 1.5mm away from case for 10 seconds		--	--	+300	℃
Vibration			IEC/EN61373 - Category 1, Grade B			
Switching Frequency *	PWM mode	3.3V/ 5V/ 6V output	--	300	--	kHz
		Others	--	270	--	
MTBF	MIL-HDBK-217F@25℃		1000	--	--	k hours

Note: *Switching frequency is measured at full load. The module reduces the switching frequency for light load (below 50%) efficiency improvement.

Mechanical Specifications

Case Material	Aluminum alloy		
Dimensions	Horizontal package (without heat sink)		25.40 × 25.40 × 11.70 mm
	Horizontal package (with heat sink)		25.40 × 25.40 × 16.20 mm
	A2S wiring package (without heat sink)		76.00 × 31.50 × 21.20 mm
	A2S wiring package (with heat sink)		76.00 × 31.50 × 25.20 mm
	A4S Din-rail package (without heat sink)		76.00 × 31.50 × 25.80 mm
	A4S Din-rail package (with heat sink)		76.00 × 31.50 × 29.80 mm
Weight	without heat sink	Horizontal package/A2S wiring package/A4S Din-rail package	15.0g/38.0g/58.0g(Typ.)
	with heat sink	Horizontal package/A2S wiring package/A4S Din-rail package	20.0g/40.0g/60.0g(Typ.)
Cooling method	Free air convection		

Electromagnetic Compatibility (EMC)

Emissions	CE	CISPR32/EN55032	CLASS B (see Fig.3-② for recommended circuit)	
	RE	CISPR32/EN55032	CLASS B (see Fig.3-② for recommended circuit)	
Immunity	ESD	IEC/EN61000-4-2	Contact ±6kV, Air ±8kV	perf. Criteria B
	RS	IEC/EN61000-4-3	10V/m	perf. Criteria A
	EFT	IEC/EN61000-4-4	±2kV (see Fig.3-① for recommended circuit)	perf. Criteria A
	Surge	IEC/EN61000-4-5	line to line ±2kV (see Fig.3-①for recommended circuit)	perf. Criteria B
	CS	IEC/EN61000-4-6	3 Vr.m.s	perf. Criteria A

Electromagnetic Compatibility (EMC) (EN50155)

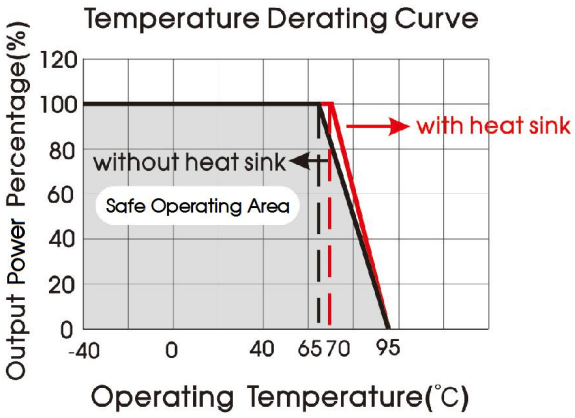
Emissions	CE	EN50121-3-2	150kHz-500kHz 99dBuV (see Fig.3-② for recommended circuit)	
		EN55016-2-1	500kHz-30MHz 93dBuV (see Fig.3-② for recommended circuit)	
	RE	EN50121-3-2	30MHz-230MHz 40dBuV/m at 10m (see Fig.3-② for recommended circuit)	
		EN55016-2-1	230MHz-1GHz 47dBuV/m at 10m (see Fig.3-② for recommended circuit)	
Immunity	ESD	EN50121-3-2	Contact ±6kV/Air ±8kV	perf. Criteria A
	RS	EN50121-3-2	20V/m	perf. Criteria A

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	EFT	EN50121-3-2 ±2kV 5/50ns 5kHz (see Fig.3-① for recommended circuit)	perf. Criteria A
	Surge	EN50121-3-2 line to line ±1kV (42Ω, 0.5μF) (see Fig.3-① for recommended circuit)	perf. Criteria A
	CS	EN50121-3-2 0.15MHz-80MHz 10V r.m.s	perf. Criteria A

Typical Characteristic Curves

Nominal input voltage, 3.3V, 5V, 6V output



Nominal input voltage, 12V, 15V, 24V output

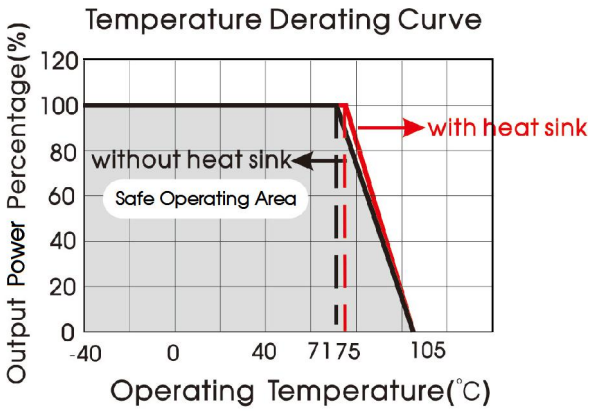
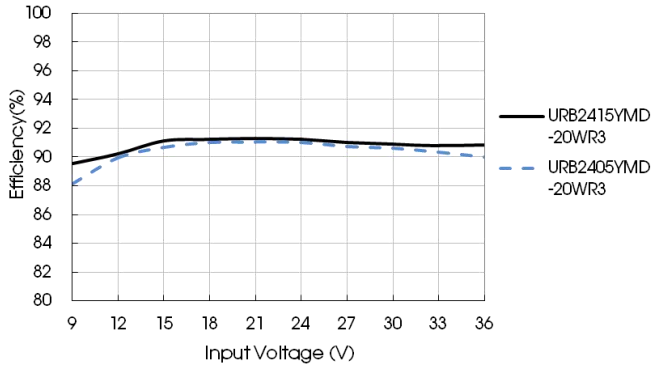
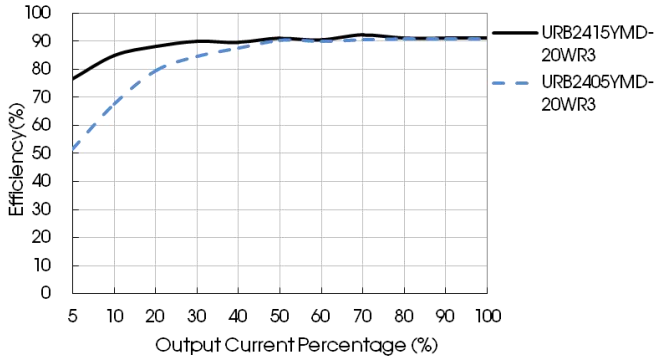


Fig. 1

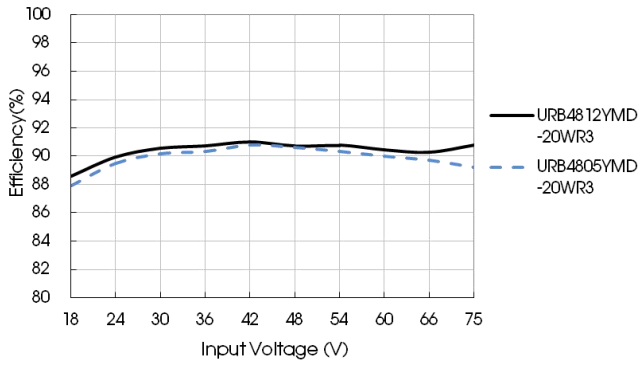
Efficiency Vs Input Voltage (Full Load)



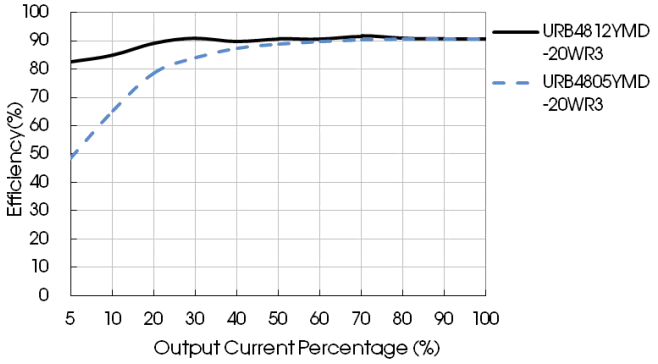
Efficiency Vs Output Load(Vin=24V)



Efficiency Vs Input Voltage (Full Load)



Efficiency Vs Output Load(Vin=48V)



Design Reference

1. Typical application

All DC-DC converters of this series are tested before delivery using the recommended circuit shown in Fig. 2.
Input and/or output ripple can be further reduced by appropriately increasing the input & output capacitor values Cin and Cout and/or by selecting capacitors with a low ESR (equivalent series resistance). Also make sure that the capacitance is not exceeding the specified max. capacitive load value of the product.

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Fig. 2

Vin (VDC)	Vout (VDC)	Cin	Cout
24	3.3/5/6	100μF/50V	100μF/16V
	12/15		100μF/25V
	24		47μF/50V
48	3.3/5	100μF/100V	100μF/16V
	12/15		100μF/25V
	24		47μF/50V

2. EMC compliance circuit

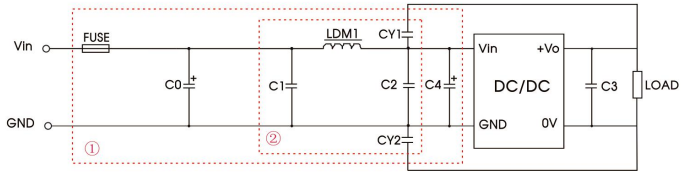


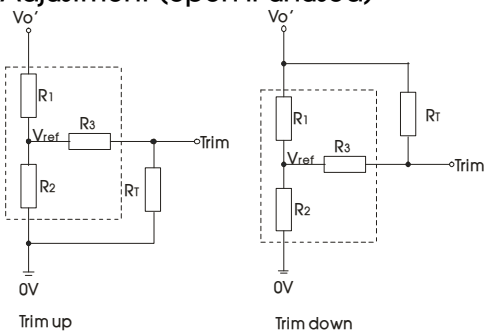
Fig. 3

Notes: We use Part ① in Fig. 3 for Immunity tests and Part ② for Emissions test. Selecting based on needs.

Parameter description:

Model	Vin: 24VDC	Vin: 48VDC
FUSE	Select fuse value according to actual input current	
C0, C4	330μF/50V	330μF/100V
C1, C2	4.7μF/50V	4.7μF/100V
C3	Refer to the Cout in Fig.2	
LDM1	2.2μH/4A	2.2μH/2A
CY1, CY2	1nF/2kV	

3. Trim Function for Output Voltage Adjustment (open if unused)



TRIM resistor connection (dashed line shows internal resistor network)

Calculating Trim resistor values:

$$\text{up: } R_T = \frac{aR_2}{R_2 - a} - R_3 \qquad a = \frac{V_{ref}}{V_o' - V_{ref}} \cdot R_1$$

$$\text{down: } R_T = \frac{aR_1}{R_1 - a} - R_3 \qquad a = \frac{V_o' - V_{ref}}{V_{ref}} \cdot R_2$$

R_T = Trim Resistor value;
a = self-defined parameter;
Vo' = desired output voltage

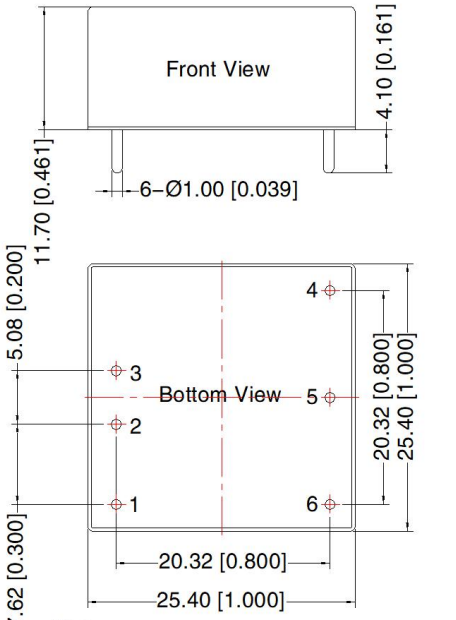
Vout(V)	R1(kΩ)	R2(kΩ)	R3(kΩ)	Vref(V)
3.3	4.775	2.87	15	1.25
5	2.894	2.87	10	2.5
6	4.064	2.87	10	2.5
12	11.000	2.87	17.4	2.5
15	14.494	2.87	17.4	2.5
24	24.872	2.87	20	2.5

4. The products do not support parallel connection of their output

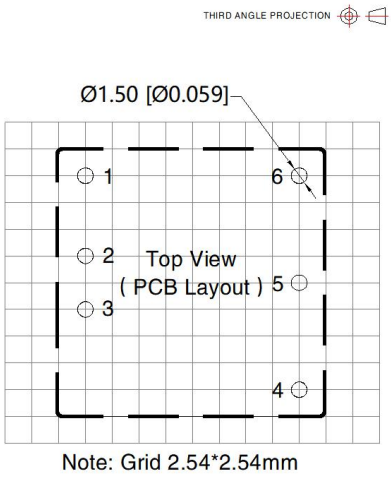
5. For additional information please refer to DC-DC converter application notes on www.mornsun-power.com

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Horizontal Package (without heat sink) Dimensions and Recommended Layout



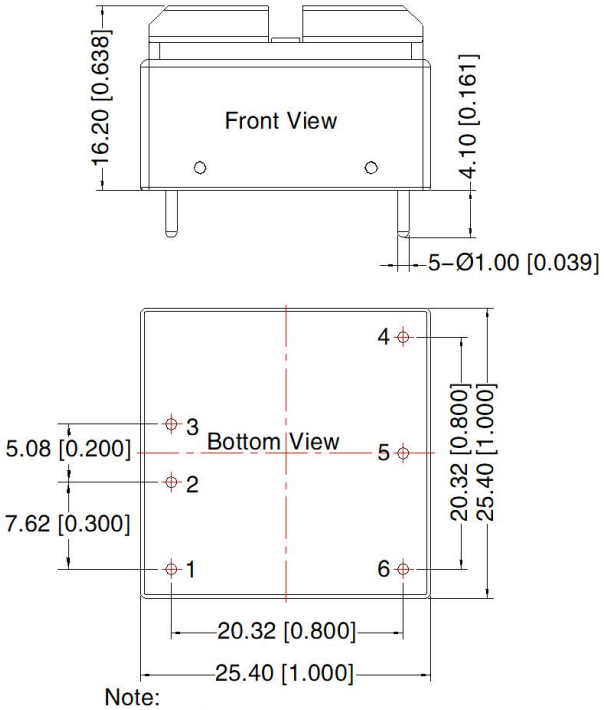
Note:
Unit: mm[inch]
PIN1/2/3/4/5/6: ϕ 1.0mm
Pin diameter tolerances: ± 0.10[± 0.004]
General tolerances: ± 0.50[± 0.020]



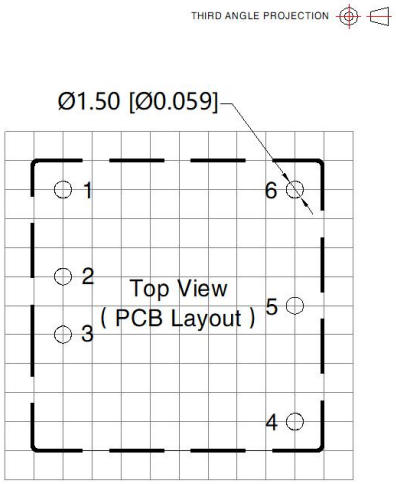
Note: Grid 2.54*2.54mm

Pin-Out	
Pin	Mark
1	Ctrl
2	GND
3	Vin
4	+Vo
5	Trim
6	0V

Horizontal Package (with heat sink) Dimensions



Note:
Unit: mm[inch]
PIN1/2/3/4/5/6: ϕ 1.0mm
Pin diameter tolerances: ± 0.10[± 0.004]
General tolerances: ± 0.50[± 0.020]



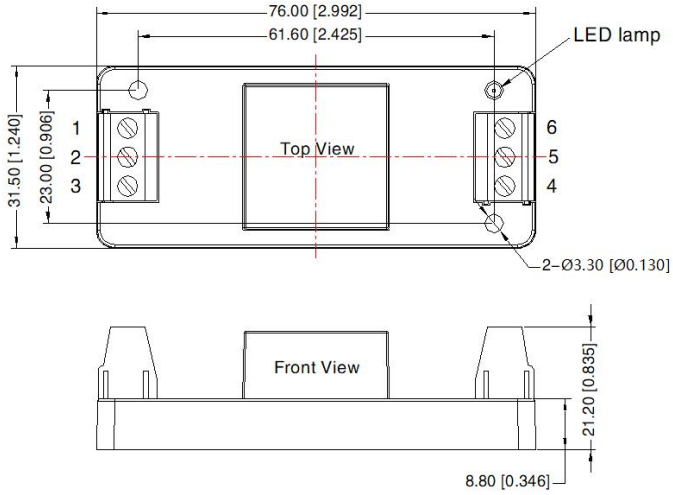
Note: Grid 2.54*2.54mm

Pin-Out	
Pin	Mark
1	Ctrl
2	GND
3	Vin
4	+Vo
5	Trim
6	0V

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URB_YMD-20WR3A2S Dimensions

THIRD ANGLE PROJECTION

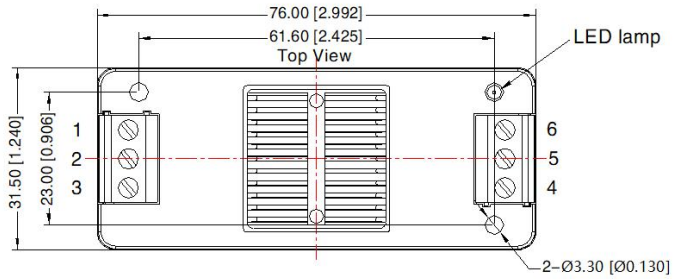


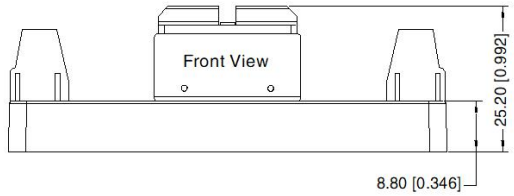
Pin-Out						
Pin	1	2	3	4	5	6
Mark	Ctrl	GND	Vin	+Vo	Trim	0V

Note:
Unit: mm[inch]
Wire range: 24-12 AWG
Tightening torque: Max 0.4 N • m
General tolerances: ± 1.00[± 0.039]

URB_YMD-20WHR3A2S (with heat sink) Dimensions

THIRD ANGLE PROJECTION





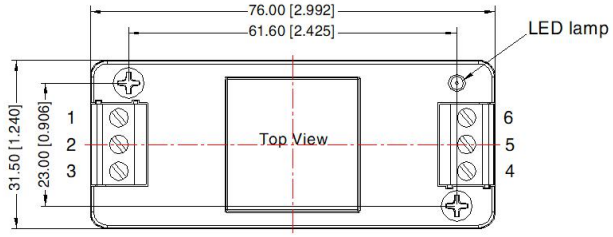
Pin-Out						
Pin	1	2	3	4	5	6
Mark	Ctrl	GND	Vin	+Vo	Trim	0V

Note:
Unit: mm[inch]
Wire range: 24-12 AWG
Tightening torque: Max 0.4 N • m
General tolerances: ± 1.00[± 0.039]

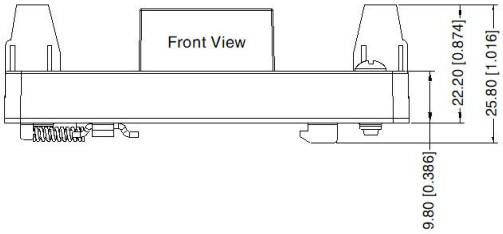
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URB_YMD-20WR3A4S Dimensions

THIRD ANGLE PROJECTION



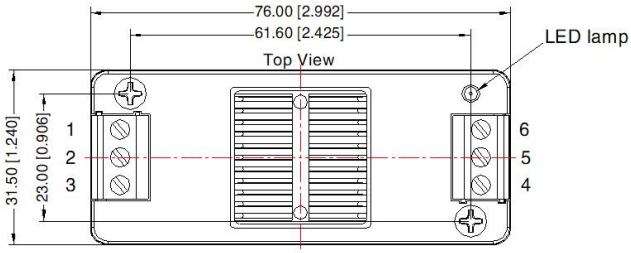
Pin–Out						
Pin	1	2	3	4	5	6
Mark	Ctrl	GND	Vin	+Vo	Trim	0V



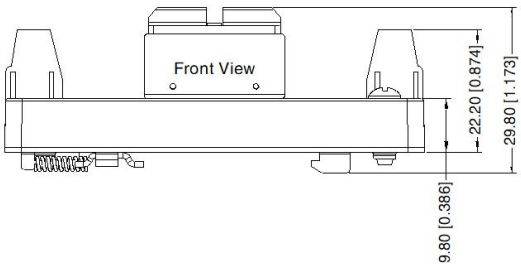
Note:
Unit: mm[inch]
Mounting rail: TS35
Wire range: 24–12 AWG
Tightening torque: Max 0.4 N • m
General tolerances: ± 1.00[± 0.039]

URB_YMD-20WHR3A4S(with heat sink) Dimensions

THIRD ANGLE PROJECTION



Pin–Out						
Pin	1	2	3	4	5	6
Mark	Ctrl	GND	Vin	+Vo	Trim	0V



Note:
Unit: mm[inch]
Mounting rail: TS35
Wire range: 24–12 AWG
Tightening torque: Max 0.4 N • m
General tolerances: ± 1.00[± 0.039]

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Note:

1. For additional information on Product Packaging please refer to www.mornsun-power.com. Packaging bag number: 58210003 (DIP), 58200048 (with heat sink), 58220022 (A2S/A4S package);
2. The maximum capacitive load offered were tested at input voltage range and full load;
3. Unless otherwise specified, parameters in this datasheet were measured under the conditions of Ta=25℃, humidity<75%RH with nominal input voltage and rated output load;
4. All index testing methods in this datasheet are based on company corporate standards;
5. We can provide product customization service, please contact our technicians directly for specific information;
6. Products are related to laws and regulations: see "Features" and "EMC";
7. Our products shall be classified according to ISO14001 and related environmental laws and regulations, and shall be handled by qualified units.

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